

Week 3 — In-Class Exercise

Transforms by hand • reallocate by hand • pick the tool

Applied Data Science and AI in Pharma Commercial • Columbia University • Fall 2026

Groups of 3-4 • 20 minutes • one group presents each part in the debrief

Handout only — write on it. Calculators or a phone are fine; no notebook needed.

Why by hand. Tonight's notebook did all of this in code. Doing it once on paper is how you learn to check a model's output at a glance — and how you explain it to a Brand Lead who will never open the notebook.

Part A • Adstock and Hill by hand (6 min)

One channel, six weeks of spend (\$k). Geometric adstock with $\delta = 0.5$; Hill saturation with $K = 300$ (\$k, adstocked) and $s = 1$; the channel's maximum weekly contribution is $\beta = 100$ TRx.

Week	1	2	3	4	5	6
Spend (\$k)	0	400	400	0	0	200
Adstock $a_t = x_t + \delta \cdot a_{t-1}$						
Hill $h = a/(a + K)$						
Contribution = $\beta \cdot h$ (TRx)						

A1. Fill in the table and total the six-week contribution.

A2. You may move one week's entire spend into another week (same total). Which move raises the six-week total the most, and why? What does the answer say about fighting vs. spreading spend for a channel with $s = 1$?

A3. Would your answer to A2 change if s were 2.5? One sentence.

Part B • Reallocate by hand (8 min)

Three channels, annual figures. The table gives the model's estimated TRx contribution (thousands) at 60 / 80 / 100 / 120 / 140% of current spend. Contribution per TRx = \$250. Interpolate linearly between columns.

Channel	Current spend \$M	60%	80%	100% (current)	120%	140%
HCP detailing	9.0	52	60	66	70	73
Digital	5.0	30	44	55	64	70
DTC TV	10.0	34	38	41	43	44.5

B1. For each channel compute average ROI at current spend, and marginal ROI going up (100% → 120%) and going down (100% → 80%). Rank the channels on average ROI and on upward marginal ROI. Which channel changes rank, and why?

B2. Move \$2M from the channel with the lowest upward marginal ROI to the channel with the highest. Recompute total TRx. What is the gain in TRx and in contribution dollars?

B3. The CFO asks instead for \$4M of savings with the smallest TRx loss. Using only 20% and 40% cuts, which combination do you propose, what does it cost in TRx and contribution, and what is the net effect on the P&L?

B4. Name one real-world constraint that could make your B2 or B3 answer infeasible in the next quarter.

Part C • Pick the tool (6 min)

Six situations a GLUCOREN brand team faced this year. For each: which tool — MMM, an experiment (which design?), digital attribution, interrupted time series, an HCP-level model, or “none, it is a data problem” — and which control or data you would need.

1. “Did the January TV flight work?” The flight ran nationally for eight weeks; TRx rose 9% and then fell back.

Tool: _____ Control / data needed: _____

2. A major PBM moved GLUCOREN to non-preferred on March 1. The brand team says “digital stopped working in Q2 — cut it.”

Tool: _____ Control / data needed: _____

3. “Which 400 HCPs should the reps call next month, and with what message?”

Tool: _____ Control / data needed: _____

4. Email open rates fell 40% after a mail-client privacy change; the agency reports that email ROI has collapsed.

Tool: _____ Control / data needed: _____

5. The brand launched eight months ago. The CFO wants ROI by channel for the annual plan.

Tool: _____ Control / data needed: _____

6. The agency's dashboard shows paid search at 12× ROI, computed from last-click conversions on the brand website.

Tool: _____ Control / data needed: _____

Debrief (5 min)

One group per part. For Part B, the presenting group states its B3 recommendation as one sentence a CFO would accept.